The Rise of Parameter Specialization for Knowledge Storage in Large Language Models

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Abstract

Over time, a growing wave of large language models from various series has been introduced to the community. Researchers are striving to maximize the performance of language models with constrained parameter sizes. However, from a microscopic perspective, there has been limited research on how to better store knowledge in model parameters, particularly within MLPs, to enable more effective utilization of this knowledge by the model. In this work, we analyze twenty publicly available open-source large language models to investigate the relationship between their strong performance and the way knowledge is stored in their corresponding MLP parameters. Our findings reveal that as language models become more advanced and demonstrate stronger knowledge capabilities, their parameters exhibit increased specialization. Specifically, parameters in the MLPs tend to be more focused on encoding similar types of knowledge. We experimentally validate that this specialized distribution of knowledge contributes to improving the efficiency of knowledge utilization in these models. Furthermore, by conducting causal training experiments, we confirm that this specialized knowledge distribution plays a critical role in improving the model's efficiency in leveraging stored knowledge.

1 Introduction

An increasing number of powerful large language models (LLMs) have emerged in recent years (Touvron et al., 2023a; Achiam et al., 2023; Groeneveld et al., 2024; Bai et al., 2023; Team, 2025), often demonstrating remarkable capabilities across various benchmarks and tests (Hendrycks et al., 2021a; Chen et al., 2021a; Cobbe et al., 2021). Thanks to the large parameter space, they have shown an exceptional ability to encode vast amounts of knowledge within their parameters, enabling superior performance on knowledge-intensive tasks (Hendrycks et al., 2021a; Zhang et al., 2023).

To understand the internal mechanism of knowledge storage, many studies have been conducted. For example, Geva et al. (2021b) interprets the MLP layers of the transformer architecture (Vaswani et al., 2017) as key-value memories, where the factual knowledge encoded in the weights is retrieved and transmitted to the output layer during inference (Geva et al., 2023; Meng et al., 2022; Yu et al., 2024). Furthermore, researchers have observed that, in the final layer of the MLP, each vector in that value matrix can act as a fundamental unit of knowledge storage (Geva et al., 2022a,b). However, there has been limited research on how to better store and compress knowledge within constrained model parameters to enable more effective utilization of that knowledge by the model.

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In this work, we investigate the relationship between language models' knowledge storage patterns and their performance. To identify parameters associated with specific knowledge concepts, we analyze consistently activated parameters in MLP layers when the model processes questions related to the particular knowledge concept. Building on the key-value interpretation of the MLP by Geva et al. (2021b), which treats the up-projection matrix as the key and the down-projection matrix as the value (*i.e.*, stored knowledge), we extract the intermediate representations between these two matrices and treat their absolute value as the activation of corresponding parameters. To support empirical analysis, we construct a new encyclopedic knowledge benchmark based on Wikipedia, covering knowledge concepts with varying frequencies. We then apply the knowledge parameter identification method to 20 open-source LLMs across a wide range of model families, enabling us to explore correlations between knowledge storage patterns and overall model performance.

Our extensive empirical analysis reveals that **stronger models exhibit higher parameter specialization for distinct knowledge**, whereas weaker models distribute knowledge more diffusely across parameters. Consequently, significantly more parameters are required to store individual knowledge in weaker models. As illustrated in Figure 1, advancing model capability correlated with improved parameter specialization for encoding knowledge: fewer parameters are allocated per knowledge concept, while each parameter governs a narrower subset of concepts.

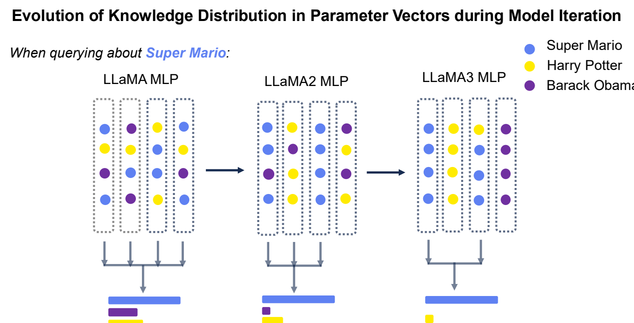

Figure 1: Evolution of knowledge distribution in model parameters during three iterations of LLaMA models. Each parameter vector corresponds to a column in the value matrix of the MLP module, as indicated by the dashed rectangles.

Motivated by this observation, we further conduct four sets of controlled experiments, each involving continued training on the Llama2-7B (Touvron et al., 2023a) and Qwen2-7B (Yang et al., 2024) models with new knowledge respectively, to validate the strong causal relationship between improved parameter specialization and enhanced performance of the models on knowledge tasks. Overall, the experiments reveal that encoding similar knowledge into the same parameter vectors better aligns with the model's internal knowledge retrieval mechanism. This approach helps the model utilize knowledge more efficiently, improves knowledge compression, and reduces hallucination generation.

Our contributions can be summarized as follows:

- To the best of our knowledge, this is the first attempt to quantify and compare the degree of parameter specialization for knowledge storage across different LLMs.
- We investigate the relationship between parameter specialization and model performance in LLMs, constructing a dedicated probing dataset for an in-depth analysis on 20 open-source LLMs. Our findings indicate that more capable LLMs exhibit greater parameter specialization.
- Through controlled training experiments, we provide empirical evidence of a causal link between increased parameter specialization and improved performance on knowledge-intensive tasks.

2 Related Work

Knowledge Storage in LLMs Studying how knowledge is stored and utilized in LLMs has been an important area in the research of LLM interpretability (Meng et al., 2022; Geva et al., 2021b; Sukhbaatar et al., 2015; Geva et al., 2023). Recent studies have shown that MLPs are the primary and crucial components for storing factual knowledge and associations in transformer-based language models (Geva et al., 2022b; Dar et al., 2023). They can be conceptualized as key-value memories (Geva et al., 2021b), where the factual knowledge encoded in the MLP weights is recalled and transmitted to the output layer during inference (Geva et al., 2023; Meng et al., 2022; Yu et al., 2024). Additionally, researchers have found that in the final layer of the MLP, each vector in the value matrix can serve as a fundamental unit for storing knowledge (Geva et al., 2022a,b). They have also verified that by directly manipulating or disrupting these parameter vectors, specific knowledge can be edited or unlearned (Hong et al., 2024a,b; Meng et al., 2022), leading to changes in the model's responses.

Knowledge Superposition in LLM Elhage et al. (2022); Olah (2023) propose the concept of Knowledge Superposition. It refers to an inevitable phenomenon in neural network models, especially large language models, during training and data memorization: since the number of data features greatly exceeds the number of parameters in the model, each parameter does not have a simple one-to-one mapping with the data features or knowledge. Neurons are often involved with multiple data features simultaneously. In our work, we treat each vector in the last layer of MLP as a basic unit for storing knowledge and investigate the superposition of knowledge within these vectors.

3 Parameter Specialization Analysis for Knowledge Storage

3.1 Preliminary

In transformer-based language models, the MLP is a crucial component for storing the model's factual knowledge, and its sub-layers can be viewed as key-value memories (Geva et al., 2021b). To be specific, the first layer[*] of MLP sublayers can be viewed as a matrix W_K formed by key vectors $\{\mathbf{k}_1, \mathbf{k}_2, \dots, \mathbf{k}_n\}$, used to capture a set of patterns in the input sequence, and ultimately outputting the coefficient scores. The second layer can be viewed as a matrix W_V formed by value vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, with each value vector containing the corresponding factual knowledge.

Formally, the output of the MLP in the transformer's ℓ-th layer, given an input hidden state $\mathbf{x}^\ell$, can be defined as:

$$\mathbf{M}^\ell = f\big(W_K^\ell \cdot \gamma(\mathbf{x}^\ell + \mathbf{A}^\ell)\big) W_V^\ell = \mathbf{m}^\ell W_V^\ell, \tag{1}$$

where $W_K^\ell, W_V^\ell \in \mathbb{R}^{n \times d}$. The function f and γ represent a non-linearity[†] and layer normalization, respectively. In the transformer's ℓ-th layer, $\mathbf{m}^\ell \in \mathbb{R}^n$ denotes the coefficient scores, and $\mathbf{A}^\ell$ represents the output of the attention component. The hidden state dimension is d, while the intermediate MLP has a dimension of n. Then, by denoting $\mathbf{v}_j^\ell$ as the j-th column (which will be called the value vector or parameter vector in the following sections) of W_V^ℓ and m_j^ℓ as the j-th element in the coefficients produced by the first layer of the MLP, we can view MLP's output $\mathbf{M}^\ell$ as a linear combination of the value vectors in W_V^ℓ, with their corresponding coefficients $\mathbf{m}^\ell$:

$$\mathbf{M}^\ell = \sum\nolimits_{j=1}^{n} m_j^\ell \mathbf{v}_j^\ell, \tag{2}$$

Finally, the hidden states at the ℓ-th layer of the language model can be defined as:

$$X^{\ell+1} = X^\ell + \mathbf{M}^\ell + \mathbf{A}^\ell, \tag{3}$$

where X^ℓ, $\mathbf{M}^\ell$ and $\mathbf{A}^\ell$ represent the hidden states, MLP's output, and the attention component's output in the transformer's ℓ-th layer, respectively. In this work, we focus on studying the impact of the MLP on the knowledge output of the hidden states.

3.2 Knowledge Vectors Masking Procedure

Referring to Eq. (2), if we aim to ablate the impact of the knowledge contained in the vectors for a particular subset S^ℓ of indices in ℓ-th layer, we can directly set the corresponding m_j^ℓ values for $j \in S^\ell$ to zero. Hence, we have:

$$\mathbf{M}^\ell_{\text{masked}} = \sum\nolimits_{\substack{j=1 \\ j \notin S^\ell}}^{n} m_j^\ell \mathbf{v}_j^\ell + \sum\nolimits_{\substack{j=1 \\ j \in S^\ell}}^{n} 0 \cdot \mathbf{v}_j^\ell = \sum\nolimits_{\substack{j=1 \\ j \notin S^\ell}}^{n} m_j^\ell \mathbf{v}_j^\ell, \tag{4}$$

[*]In most decoder-only models, such as GPT-2 (Radford et al., 2019) and GPT-J (Chen et al., 2021b), the MLP component consists of two layers, whereas in LLaMA (Touvron et al., 2023b), it comprises three layers. However, we can still regard LLaMA's first two layers collectively as the key matrices, with their output representing the coefficient scores.

[†]For brevity, the bias term is omitted.

Therefore, given a concept, when we aim to identify which specific value vectors in the model's MLPs are most closely related to the knowledge contained in that concept—while avoiding the masking of vectors associated with the model's general capabilities[‡] (Meng et al., 2022; Geva et al., 2023), i.e., determining the appropriate subset S^ℓ at each layer of the model for this concept—we will run t concept-related questions and t^* irrelevant questions on the selected model. Then we will compute the corresponding coefficients $\mathbf{m}^\ell$ and $\mathbf{m}^{*\ell}$, which are the averages of the coefficients for the concept-related questions and irrelevant questions, respectively, at each layer of the model. For details on the generation of concept-related and irrelevant questions, as well as the selection of t and t^*, please refer to §3.4. After obtaining $\mathbf{m}^\ell$ and $\mathbf{m}^{*\ell}$ at each layer, we perform the computation using the following formula:

$$\mathbf{S}^\ell = \left\{ \left| m_j^\ell - m_j^{*\ell} \right| \,\middle|\, 1 \leq j \leq n,\ m_j^\ell \in \mathbf{m}^\ell,\ m_j^{*\ell} \in \mathbf{m}^{*\ell} \right\} \tag{5}$$

Next, we will sort $\mathbf{S}^\ell$ in descending order and select the value vectors corresponding to the indices of the top k elements, which will be used as the subset S^ℓ for the masking operation. This allows us to observe and analyze the impact of masking these vectors on the model's knowledge output for certain concepts.

3.3 The Definition of Parameter Specialization

After obtaining the subset of value vectors S^ℓ that exhibit specificity to a given concept at each layer of the model, as described in §3.2, we apply the masking operation to these value vectors, as shown in Eq. (4). We then analyze its impact on the model's final outputs for the t concept-related questions and t^* irrelevant questions. By comparing the model's responses after masking with the ground truth answers, we compute the accuracy on concept-related questions, referred to as the Concept Specific Score after surgery, and the accuracy on irrelevant questions, referred to as the General Score after surgery. To quantify the degrees of specialization of the model's value vectors with respect to the concept-related knowledge, we define the **Parameter Specialization Score (PSS)**:

$$\text{PSS} \triangleq \frac{|\text{General Score after surgery} - \text{Concept Specific Score after surgery}|}{\text{General Score before surgery}}, \tag{6}$$

which is obtained by taking the absolute difference between the General Score and the Concept Specific Score after surgery, and then dividing by the model's accuracy on the entire dataset before surgery. A higher PSS indicates that the parameter vectors in the model's MLP layers exhibit a higher degree of specialization towards specific knowledge. Conversely, a lower PSS suggests more severe knowledge superposition phenomena within the parameter vectors, resulting in a lower degree of specialization.

3.4 Dataset Construction

To thoroughly investigate the parameter specialization of knowledge with different frequencies in the parameter vectors of LLMs' MLP, we introduce a dataset named SpecWiki. It includes 525 concepts selected from Wikipedia [§], a widely recognized high-quality corpus for LLM training. These concepts are categorized based on their frequency levels to ensure a diverse distribution. We then design two distinct question formats—multiple-choice questions and open-ended generation prompts—to facilitate a thorough examination of the models' knowledge storage.

Concept Selection We treat each Wikipedia item as a defining concept, typically represented by an article focused on a specific subject, indicated by its title. We focus on specific entity concepts, such as historical figures, events and locations. We began by randomly sampling 2,400 pages (a 0.01% rate) from the 2019 version of Wikipedia. Subsequently, we performed manual filtering to remove overly commonsensical or abstract concepts (such as the letter 'S' and the word 'Freedom'), ambiguous concepts (like 'Apple'), and those associated with pages under 1,000 words. Ultimately, this resulted in 525 high-quality concepts spanning specific topics like people, arts, and events.

[‡]The term "general ability" refers to the model's fundamental skills, such as processing text inputs correctly and generating coherent outputs, rather than encoding knowledge specific to a particular concept.

[§]https://en.wikipedia.org/